FINSIGHTS

DS5500 : Check-in 05

Omkar Sonar

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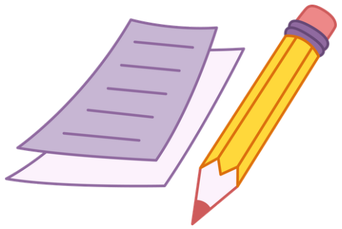
INTRODUCTION

- Conversation with EVP at Bangor Savings Bank
- Anomaly detection in transactions

PROBLEM STATEMENT



- Pivot from Anomaly detection → Customer Intelligence System POC
- Why? No fraud labels in data
- Potential use cases -
  - Complements bank rewards program (Buoy Local)
  - Local business promotions
  - Personalized cashback rewards
  - Identify customers who might be struggling

DATA AND PIPELINE

2 Data sources

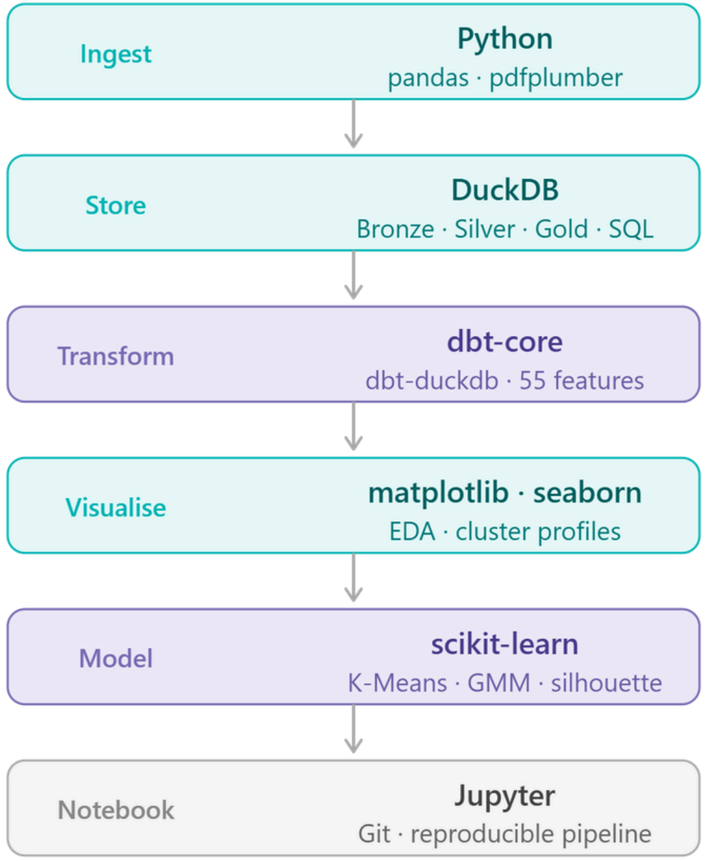
Transactions - 8.2 M rows (Nov ‘25 - Jan ’26)

Customers - 278K rows (demographics)

Custom CSV parser

Overlap : 68,585 customers

Raw Files → Bronze → Silver → Gold



FEATURE ENGINEERING

	category	transaction_count
0	Dining	1541987
1	Grocery	1513183
2	Gas	1188197
3	Retail	973580
4	Other	476853
5	Digital/Subscriptions	476337
6	Telecom/Cable	289457
7	Electronics/Computers	275133
8	Financial Services	247389
9	Entertainment/Leisure	211114
10	Healthcare	171099
11	Automotive	131616
12	Home/Hardware	130801
13	Personal Care	114138
14	Transportation	103357
15	Government/Community	95945
16	Clothing	92382
17	Pets	53901
18	Utilities	50906

customer_id	BSB008064
Individual	Y
total_spend	6434.37
transaction_count	115
avg_transaction_amount	55.95
median_transaction_amount	22.22
std_transaction_amount	73.46
max_transaction_amount	398.36
min_transaction_amount	2.99
grocery_share	0.368
dining_share	0.0
gas_share	0.0844
digital_share	0.009
electronics_share	0.0082
telecom_share	0.0937
retail_share	0.3036
healthcare_share	0.0
pets_share	0.0
home_share	0.0
automotive_share	0.0
clothing_share	0.0
entertainment_share	0.0
financial_share	0.1197
personal_care_share	0.0

transport_share	0.0
utilities_share	0.0
government_share	0.0
other_share	0.0134
weekend_share	0.4261
avg_transactions_per_week	12.2
dec_to_nov_daily_spend_ratio	1.5479
jan_to_dec_daily_spend_ratio	0.9438
unique_merchants	47
top_merchant_share	0.3051
repeat_merchant_rate	0.2128
repeat_transaction_rate	0.5913
avg_visits_per_merchant	2.45
Age	37
RelationshipYears	4
is_male	0
is_female	1
NumberActiveDDAs	2
NumberActiveTimeDeposits	0
NumberActiveLoans	1
NumberCreditCardAccts	0
has_deposit	1
has_loan	1
has_creditcard	0
has_atm	0

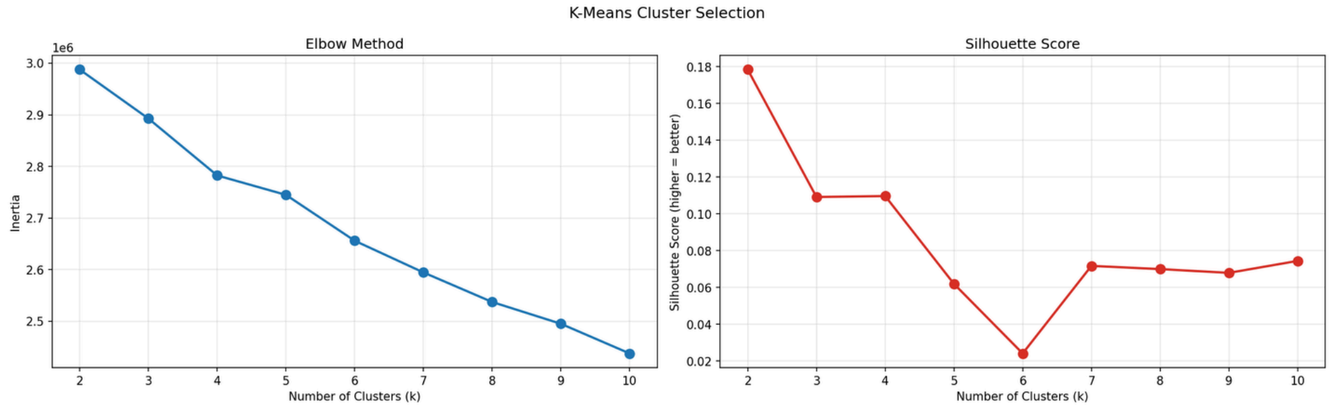
has_debit	1
is_primary	1
is_wealth	0
is_payroll	0
is_pling	0
is_able	0
dec_to_nov_avg_amount_ratio	1.3276
jan_to_dec_avg_amount_ratio	1.1616

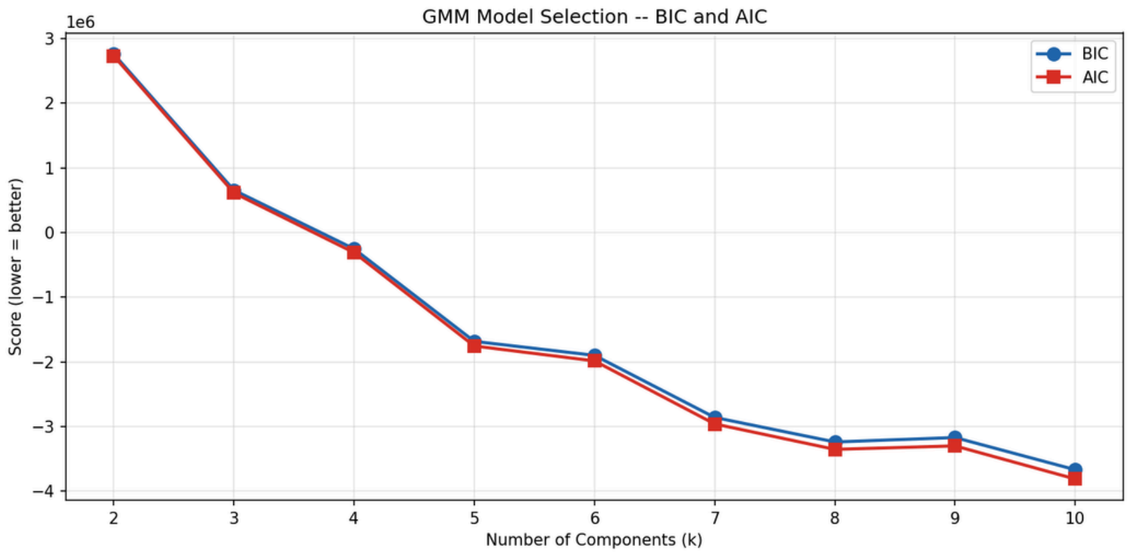
CLUSTERING METHODOLOGY

Pre-processing : Removing 163 outliers | Log transformations | Skewness adjusted | Standardization

K-Means : elbow + silhoutte

GMM : BIC + AIC





RESULT

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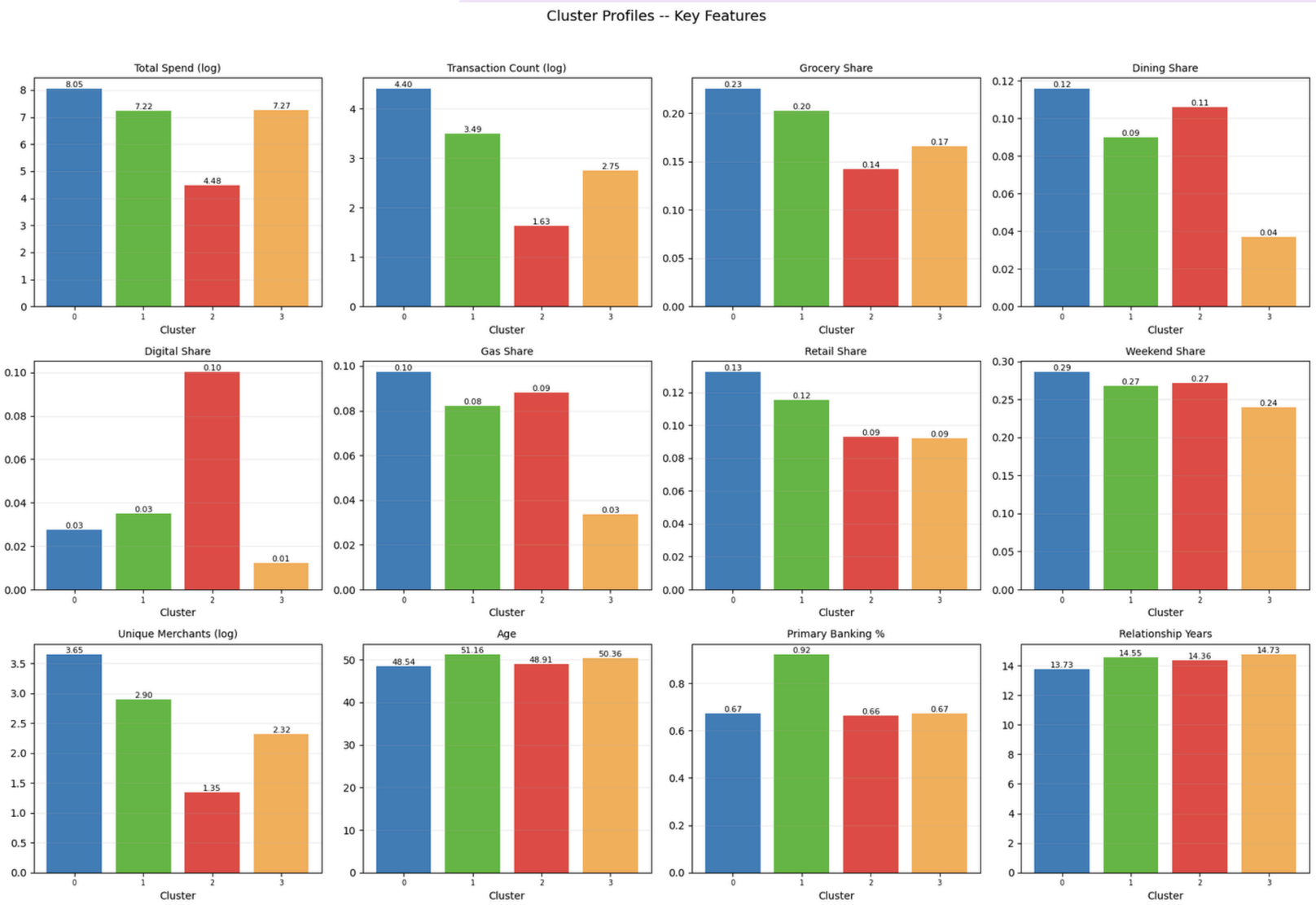
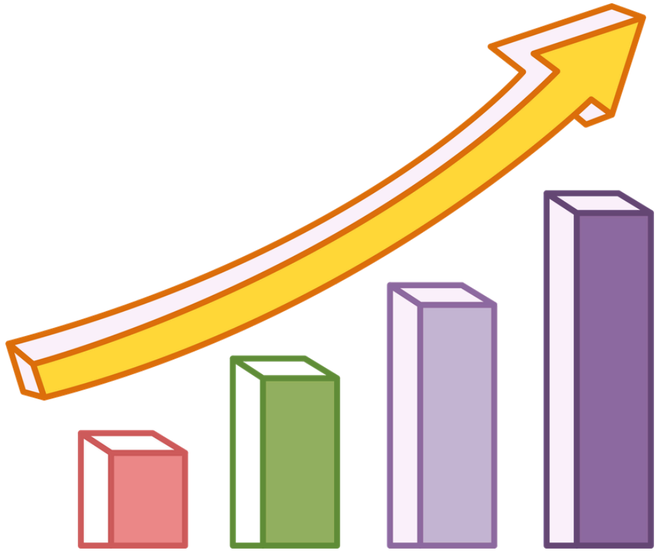
  Cluster 0: 33,453 customers (57.0%)
  Cluster 1:  1,396 customers (2.4%)
  Cluster 2:  9,970 customers (17.0%)
  Cluster 3: 13,858 customers (23.6%)
```

cluster	0	1	2	3
customer_count	33453	1396	9970	13858
avg_total_spend	8.047	7.225	4.484	7.266
avg_txn_count	4.403	3.488	1.626	2.751
avg_grocery	0.225	0.202	0.142	0.166
avg_dining	0.116	0.09	0.106	0.037
avg_gas	0.097	0.082	0.088	0.034
avg_digital	0.028	0.035	0.1	0.012
avg_retail	0.133	0.115	0.093	0.092
avg_age	48.536	51.161	48.913	50.361
avg_relationship_yrs	13.734	14.554	14.357	14.73
avg_weekend_share	0.286	0.268	0.272	0.24
avg_unique_merchants	3.646	2.896	1.347	2.323
pct_primary	0.671	0.92	0.663	0.671
pct_wealth	0.007	0.023	0.007	0.008



NEXT STEPS

1. Stakeholder Review
2. Mapping cluster segments with initial use-cases
3. Supervised classifier for segmentation of new classifiers
4. Spend tier prediction based on spending changes



THANK YOU